

Your **Assessment Tool 2** is a **Concept Mapping Assessment** for *Cryptography and Network Security (CSA51)* with **5 questions carrying 50 marks**.

You can write it in the following assessment format:

SIMATS ENGINEERING

ASSESSMENT TOOL – 2

COURSE: Cryptography and Network Security (CSA51)

Concept Mapping Assessment

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Question 1: Security Attacks, Threats, Risks and Security Goals

Concept Map

Security Attacks

↓

Threats

↓

Risks

↓

Security Goals

(Confidentiality, Integrity, Availability)

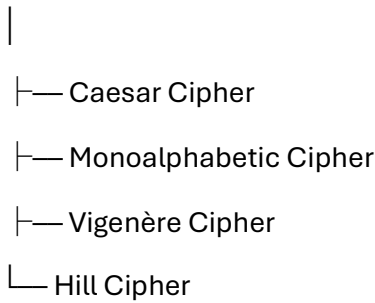
Attack → Threat → Risk → Security Goal

Attack	Threat	Risk	Security Goal	Example
Eavesdropping	Information Disclosure	Data Leakage	Confidentiality	Hacker intercepting bank passwords
Data Modification	Tampering	Incorrect Information	Integrity	Changing exam marks in a database
DoS Attack	Service Disruption	System Unavailability	Availability	Website server crash due to flooding

Question 2: Substitution Techniques

Concept Map

Substitution Techniques



Cipher	Key Usage	Complexity	Security Strength
Caesar Cipher	Single shift key	Very Low	Weak
Monoalphabetic Cipher	Fixed substitution alphabet	Low	Moderate
Vigenère Cipher	Repeating keyword	Medium	Strong
Hill Cipher	Matrix key	High	Very Strong

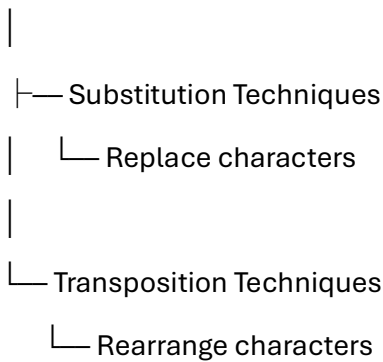
Analysis

Hill Cipher is the most secure because it uses matrix operations and multiple substitutions, making frequency analysis difficult.

Question 3: Substitution vs Transposition Techniques

Concept Map

Classical Encryption



Similarities

- Both are classical encryption methods.
- Both convert plaintext into ciphertext.
- Both require a key.

Differences

Substitution

Characters are replaced

Changes character identity

Frequency analysis possible

Transposition

Characters are rearranged

Changes character position

Pattern analysis possible

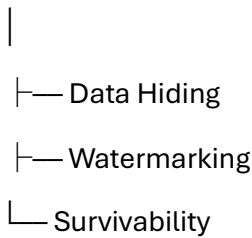
Vulnerability Analysis

Substitution techniques are more vulnerable because letter frequency can reveal the plaintext.

Question 4: Steganography

Concept Map

Steganography



Concept	Contribution	Real-World Application
Data Hiding	Conceals secret messages	Hidden military communication
Watermarking	Copyright protection	Digital image ownership
Survivability	Resists attacks and modifications	Secure cloud storage

Question 5: Number Theory Concepts in Cryptography

Concept Map

Extended Euclid Algorithm



Modular Inverse



Euler Phi Function



Euler's Theorem



Fermat's Little Theorem



Public Key Cryptography (RSA)

Encryption/Decryption Role

Concept	Role
Extended Euclid Algorithm	Finds GCD and modular inverse
Modular Inverse	Calculates private key
Euler Phi Function	Generates RSA keys
Euler's Theorem	Basis of RSA encryption
Fermat's Little Theorem	Simplifies modular exponentiation

Analysis

Euler's Phi Function is the most critical concept for public key cryptography
because RSA key generation directly depends on calculating $\phi(n)$.

This format is simple, neat, and suitable to write directly in your assessment notebook.